Aijia (Annie) Yang

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EDUCATION

University of California San Diego, CA **Expected March 2028**
B.S. Mechanical Engineering with Specialization in Controls & Robotics
B.S. Mathematics-Applied Science (Mechanical Engineering & Computer Science Focus)

- The Safe Autonomous Systems Lab, Apple Next-Gen Innovator Program, Tau Beta Pi Engineering Honors Society, Triton Robotics, Society of Women Engineers (SWE), MAE Scholars, Association for Computing Machinery (ACM)
- Courses: Linear Circuits, Data Structures & Algorithms, Probability, Object-Oriented Programming, Vector Calculus, Linear Algebra, Robotics Motion & Planning, Signals & Systems (Fall 2026)

EXPERIENCE

Robotics Research Intern **June 2026 – September 2026**
Toyota Research Institute (https://www.tri.global/) *Los Altos, CA*

- First-author paper at the 3rd Workshop on Dexterous Manipulation, Conference on Robot Learning (CoRL 2026).
- Integrated tactile sensing into robotic control pipelines to support sensor-informed policy training and evaluation.
- Leveraged NVIDIA Isaac Sim to simulate, test, and validate learning-based robotic control policies.
- Developed internal tools for robotics experiment management, data visualization, and policy evaluation.
- Built data analysis and testing frameworks to assess policy performance, reliability, and learning workflows.

Research Assistant **January 2025 - Present**
The Safe Autonomous Systems Lab (https://sylviaherbert.com/) *San Diego, CA*

- Co-authored IEEE International Conference on Robotics and Automation (ICRA 2026) paper "MADR: MPC-guided Adversarial DeepReach" (https://land-dev.github.io/madr/) on safe learning-based control.
- Developed ROS2 and Python infrastructure supporting deployment and validation of robotics hardware experiments.
- Extended infrastructure from single-agent to multi-agent robot interactions to test safe control policies using model-predictive control, deep learning, and reachability analysis.
- Conducted hardware validation on Crazyflie UAVs, TurtleBots, and humanoids to evaluate safe control algorithms.

Mechanical Engineer **September 2024 - Present**
Triton Robotics (https://tritonrobotics.org/) *San Diego, CA*

- Designed and fabricated mechanical subsystems for autonomous competition robots, integrating electronics protection, manufacturability, and field reliability.
- Manufactured critical robot components using laser cutting, 3D printing, and waterjet for the RoboMaster North America Competition.
- Engineered battery shields and exterior armor for 3v3 "Infantry" robot, enhancing durability and on-field performance.

PROJECTS

Interstellar Robotics **September 2025 – December 2025**
UC San Diego Mechanical and Aerospace Engineering Department *San Diego, CA*

- Designed, fabricated, and tested a fully functional competition robot capable of scoring all game elements (asteroids, meteors, and comets) in a class-wide robotics competition. Received the Best Presentation Award.
- Analyzed and integrated mechanical and robotic components to optimize performance, reliability, and scoring efficiency under competition constraints.

Earthquake Wonders: High School Investigation in Mortise-Tenon Structure **September 2022 – April 2023**
Charles Wright Academy (publication: http://dx.doi.org/10.54254/2753-8818/9/20240718) *Tacoma, WA*

- Investigated how mortise-tenon structures reduce earthquake vibration by generating frictional damping.
- Proved that mortise-tenon structures reduce vibration by 11.0% compared to screw and glue structures.
- Developed and validated an innovative low-cost sensor system with accessible tools.

SKILLS/AWARDS

Programming: Python, Java, MATLAB, C#, Linux, Docker, Git
Robotics: ROS2, Isaac Sim, Motion Planning, Robot Learning, Controls
Engineering: Onshape, Fusion 360, AutoCAD, SolidWorks, FEA, 3D Printing, Laser Cutting, Design for Manufacturing
Awards/Activities: Tau Beta Pi Vice President, AIME, WA Science & Eng Fair 2nd place, 30th ISSDC global champion